

Finding Time Complexity of Algorithms

Problem 1: Finding Complexity using Counter Method

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:(penalty regime: 0 %)

Code

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int c=0;
```

```

int s=1;

c++;

int i=1;

c++;

while(s<=n)
{
    c++;

    i++;

    c++;

    s+=i;

    c++;

}

c++;

printf("%d",c);

return 0;

}

```

Input	Expected	Got	
	9	12	12
	4	9	9

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Problem 2: Finding Complexity using Counter method

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:(penalty regime: 0 %)

Code

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int c=0;
```

```
    if(n==1)
```

```
    {
```

```
        c++;
```

```
    }
```

```
    else
```

```
    {
```

```
        c++;
```

```

for(int i=1;i<=n;i++)
{
    c++;
    for(int j=1;j<=n;j++)
    {
        c++;
        c++;
        c++;
        break;
    }
    c++;
}
c++;
}
printf("%d",c);
return 0;
}

```

Input	Expected	Got	
	2	12	12
	1000	5002	5002
	143	717	717

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Problem 3: Finding Complexity using Counter Method

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
    {  
        for (i = 1; i <= num;++i)  
        {  
            if (num % i== 0)  
            {  
                printf("%d ", i);  
            }  
        }  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Code

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int num;
```

```
    scanf("%d",&num);
```

```
    int c=0;
```

```
    for (int i = 1; i <= num;++i)
```

```
    {
```

```
        c++;
```

```
    if (num % i== 0)
```

```
    {
```

```
        c++;
```

```
    }
```

```
    c++;
```

```
}
```

```

c++;

printf("%d",c);

return 0;

}

```

Input	Expected	Got	
	12	31	31
	25	54	54
	4	12	12

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

[Finish review](#)

Problem 4: Finding Complexity using Counter Method

Question 1

Correct

Mark 1.00 out of 1.00

[Flag question](#)

Question text

Convert the following algorithm into a program and find its time complexity using counter method.

```

void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Code

```
#include<stdio.h>

int main()
{
    int n;

    scanf("%d",&n);

    int count=0;

    int c=0;

    c++;

    for(int i=n/2; i<n; i++)
    {
        c++;

        for(int j=1; j<n; j = 2 * j)
        {
            c++;

            for(int k=1; k<n; k = k * 2)
            {
                c++;

                count++;

                c++;
            }

            c++;
        }

        c++;
    }

    c++;

    printf("%d",c);

    return 0;
}
```

Input	Expected	Got	
	4	30	30
	10	212	212

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

Problem 5: Finding Complexity using counter method

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Code

```
#include<stdio.h>
```

```
int main()
```

```

{
    int n;
    scanf("%d",&n);
    int c=0;
    int rev = 0, remainder;
    c++;
    while (n != 0)
    {
        c++;
        remainder = n % 10;
        c++;
        rev = rev * 10 + remainder;
        c++;
        n/= 10;
        c++;
    }
    c++;
    c++;
    printf("%d",c);
    return 0;
}

```

Input	Expected	Got	
	12	11	11
	1234	19	19

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.